

ALLEGRETTI GIOVANNI

Senior Software Engineer

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🌐 [LinkedIn](#)

📍 Rome (IT)

SUMMARY

Senior C++ Software Engineer with 5+ years of experience in performance-critical development, Qt/QML desktop applications, and computer vision. Strong background in software architecture design and DevOps integration. Passionate about robotics and embedded systems.

EXPERIENCE

DMA Accurate & Reliable - Software Engineer

04/2023 – Present, Turin (IT)

- Lead developer of core components of the real-time video inspection system, responsible for architecture refactoring, API design, and performance-critical modules supporting neural network-based analysis pipelines and real-time processing.
- Designed and developed full-stack desktop applications with Qt/QML (C++ backend), including internal tools and customer-facing applications for railway infrastructure analysis.
- Led the evolution of the catenary wear analysis system, delivering significant backend enhancements and a fully redesigned internal UI.
- Contributed to DevOps integration efforts, including Docker containerization of core libraries, GitLab CI/CD pipeline setup, and workflow orchestration.

ALTEN (consultant at Leonardo Aircraft) - Software Engineer

07/2020 – 03/2023, Turin (IT)

- Designed and developed HMI applications in C++ and Qt for avionics mission simulation, enabling real-time monitoring of cockpit inputs and injection of simulated failures.
- Developed and maintained a flight and mission simulation model in C++, focusing on reliability, performance, and system integration.

ALTEC - Aerospace Logistics Technology Engineering Company - Software Engineer

03/2019 – 07/2019, Turin (IT)

- Designed, developed, and tested a C++ software module for planetary exploration robots, computing terrain-adaptive kinematics for a 6-DOF robotic arm. Integrated into an existing ROS/Gazebo framework.

PROJECTS AND COLLABORATIONS

Robotic arm industrial application - Cognitive Robotics

Università Degli Studi di Salerno

2019, Fisciano (IT)

- Designed and developed a cognitive robotics system for object recognition and manipulation using computer vision and neural networks.

Autonomous driving - Computer Vision

Università Degli Studi di Salerno

2019, Fisciano (IT)

- Developed an autonomous rover system for navigation in a controlled environment.

Rover driving - Embedded systems

Università Degli Studi di Salerno

2018, Fisciano (IT)

- Designed and implemented a PID controller and piloting firmware for manual and semi-automatic driving.

EDUCATION

Sentiment analysis - Computer vision

Master thesis Erasmus Program - University of Groningen

2019, Groningen (NL)

- Automatic Facial Expression Recognition in an uncontrolled environment.

Master's Degree in Computer Engineering

Università Degli Studi di Salerno

GPA **110/110**
2020, Fisciano (IT)

Bachelor's Degree in Computer Engineering

Università Degli Studi di Salerno

GPA **95/110**
2017, Fisciano (IT)

LANGUAGES

Italian

Native



English

Advanced



TECHNICAL SKILLS

Core

C++

Qt/QML

Computer Vision

Multithreading

Software Architecture

Design Patterns

Tools & DevOps

Git

GitLab

CI/CD

Docker

CMake

GTest

Valgrind

Kestra

Libraries & Frameworks

OpenCV

Boost

ZMQ

ROS

TensorFlow

Platforms

Linux

Windows

Embedded (ESP32,

STM32)

Nvidia Jetson

Other Languages

Python

C

Ada

MATLAB

Bash

Methodologies

Agile/Scrum

PERSONAL TRAITS

Problem Solving Teamwork

Critical Thinking Attention to Detail

Continuous Learning